EVERY NON-ASSOCIATIVE ALGEBRA EMBEDS INTO AN ASSOCIATIVE ALGEBRA THROUGH A DERIVATION

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Allowed standard prerequisites (editorial).

- The free associative algebra generated by an enriched set with an induced well-order. standard construction used at proof lines 827-832. Supplies the ambient algebra in which the embedding is constructed, together with the order used to analyse the products.
- The rewriting of products of ordered monomials through the derivation induced on the free algebra. derived inside the proof at proof lines 832-859. Shows that the symmetrized products of the images of the elements of the given algebra reproduce its multiplication.

Proposition 3.7. *For every non-associative algebra V with a multiplication $\nu\colon V\otimes V\to V$ there exists an associative algebra $(A,\cdot)$ with a derivation d such that $V\subseteq A$ and $\nu(u,v)=d(u)\cdot v+v\cdot d(u)$ for all $u,v\in V$.*

Proof. Let us choose a linear basis B of V equipped with an arbitrary total order $\leq$ such that $(B,\leq)$ is a well-ordered set. Then define F to be the free associative algebra generated by $B^{(\omega)}$. Induce the order $\leq$ on $B^{(\omega)}$ by the following rule:

$$a^{(n)}\leq b^{(m)} \iff (n,a)\leq(m,b) \text{ lexicographically},$$

and expand it to the words in $B^{(\omega)}$ by the deg-lex rule (first by length, then lexicographically).

Consider the set of defining relations

$$S=\left\{a^{(n)}b+\sum_{s\geq 1}\binom{n-1}{s}\big(a^{(n-s)}b^{(s)}+b^{(s)}a^{(n-s)}\big)+ba^{(n)}-\nu(a,b)^{(n-1)}\mid a,b\in B,n\geq 1\right\}.$$

All relations in the set S are obtained from $a'b+ba'-\nu(a,b)$ by formal derivation $d\colon x^{(s)}\to x^{(s+1)}$, $x^{(s)}\in B^{(\omega)}$. Hence, $A=F/(S)$ is a differential associative

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algebra, and the map $\varphi\colon V \to A^{(+)}$, $\varphi(v) = v + (S)$, $v \in V$, preserves the operation, i.e., $\varphi(\nu(u,v)) = d(\varphi(u)) \cdot \varphi(v) + \varphi(v) \cdot d(\varphi(u))$ for all $u, v \in V$.

The principal parts of $f \in S$ relative to the order $\leq$ are $a^{(n)}b$, $a, b \in B$, $n \geq 1$. These words have no compositions of inclusion or intersection, hence, S is a Gröbner–Shirshov basis in F and the images of all variables from B are linearly independent in A since they are S-reduced (see, e.g., [1] for the definitions). Therefore, $\varphi\colon V \to A^{(+)}$ is the desired embedding. □

References

[1] Bokut L.A., Chen Y., Gröbner–Shirshov bases and their calculation, *Bull. Math. Sci.* **4** (2014), 325–395, arXiv:1303.5366.